Q.1

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded by a constant value. What is the significance of this boundedness?

A.

B. It ensures the existence of the Laplace transform.

C. It guarantees the continuity of the function.

D. It guarantees the differentiability of the function.

It ensures the periodicity of the function.

Q.2 Given the piecewise function: f(t) = \begin{cases} 0, & t < 2 \\ 5, & 2 \leq t < 4 \\ 0, & t \geq 4 \end{cases} Which of the following correctly represents f(t) using unit step functions?

A. f(t)=5□[u(t−2)−u(t−4)]

B. f(t)=5□[u(t)−u(t−2)]

C. f(t)=5□u(t−2)

D. f(t)=5□[u(t−2)+u(t−4)]

Q.3 What is the Laplace transform of the first derivative f'(t) of a function f(t)?

A. sF(s)−f(0)

B. sF(s)+f(0)

C. (1/s)F(s)−f(0)

D. (1/s)F(s)+f(0)

Q.4 Which of the following correctly describes the relationship between the Dirac delta function δ(t) and the unit step function u(t)?

A. u(t)=δ(t)□ t

B. \ u(t)= {d}{dt} (t)

C. (t) = {d}{dt} u(t)

D. δ(t)=u(t)□t

Q.5

What is the Laplace transform of the derivative of a function $f(t)$ denoted by $f\prime(t)$?

A.

$$s\mathcal{L}\{f(t)\}$$

B.

$$\mathcal{L}\{f(t)\}$$

C.

$$s\mathcal{L}\{f(t)\} - f(0)$$

D.

$$f(0) - s\mathcal{L}\{f(t)\}$$

Q.6 Which of the following correctly represents the convolution of two continuous-time signals x(t) and h(t)?

A. y(t) = _{0}^{t} x() \, h(t-) \, d

B. y(t) = _{-}^{} x() \, h() \, d

C. y(t) = _{0}^{t} x(-1) \, h(1-) \, d

D. y(t) = _{0}^{} x() \, h(1-) \, d

Q.7 Given: f(t)=(t−3)2□u(t−3) What is the Laplace transform L{f(t)}?

A. 2/ s3

B. e−3s. 2/s3

C. e-s s-2

D. 2e3s / s3

Q.8

If $f(t) = e^t$, then the Laplace transform of $f'(t)$ is:

A.

$$\frac{s^2}{s-1}$$

B.

$$\frac{s}{s+1}$$

C.

$$\frac{s}{s-1}$$

D.

None of other choices

Q.9

The condition $|f(t)e^{-kt}| \leq M$ ensures that the function is bounded for all $t \geq 0$. What does M represent in this condition?

A.

Minimum value of the function

B.

Maximum value of the function

C.

Upper bound on the function

D.

Lower bound on the function

Q.10

If $f(t) = \sin t$, then the Laplace transform of $f'(t)$ is:

A.

$\mathcal{L}(f\prime(t)) = \frac{-s}{s^2+1}$

B.

$\mathcal{L}(f\prime(t)) = \frac{s^2}{s^2+1}$

C.

$\mathcal{L}(f\prime(t)) = \frac{s}{s^2+1}$

D.

$\mathcal{L}(f\prime(t)) = \frac{-s^2}{s^2+1}$